(4) What is the chance that
(i) non-leap year (ii) leap year should have fifty three Sundays?

(5) Eight coins are tossed once, find the probability of getting
(i) exactly two tails (ii) atleast two tails (iii) atmost two tails

(6) An integer is chosen at random from the first 100 positive integers. What is the probability that the integer chosen is a prime or multiple of 8?

(7) A bag contains 7 red and 4 black balls, 3 balls are drawn at random.
Find the probability that (i) all are red (ii) one red and 2 black.

(8) A cricket club has 16 members, of whom only 5 can bowl. What is the probability that in a team of 11 members at least 3 bowlers are selected?

(9) (i) The odds that the event A occurs is 5 to 7, find $P(A)$.

(ii) Suppose $P(B) = \frac{2}{5}$. Express the odds that the event B occurs.

12.5 Some basic Theorems on Probability

The problems solved in the last sections are related to mutually exclusive events. So we have used the formula $P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$. But when the events are mutually inclusive, the additivity axioms counts $(A \cap B)$ twice. We have a separate formula for the events when they are mutually inclusive.

In the development of probability theory, all the results are derived directly or indirectly using only the axioms of probability. Here we derive some of the basic important theorems on probability.

Theorem 12.3

The probability of the impossible event is zero. That is,

$$\boxed{P(\varnothing) = 0}$$

Proof

Impossible event contains no sample point.

$$\begin{aligned} \text{Therefore, } S \cup \varnothing &= S \\ P(S \cup \varnothing) &= P(S) \\ P(S) + P(\varnothing) &= P(S) \qquad \text{(since } S \text{ and } \varnothing \text{ are mutually exclusive)} \\ P(\varnothing) &= 0 \end{aligned}$$

Example 12.12

Find the probability of getting the number 7, when a usual die is rolled.

Solution

The event of getting 7 is an impossible event. Therefore, $P(\text{getting } 7) = 0$

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